SAMPLE TEST

SECTION 1 Time—30 minutes, 45 Questions. For each question in this section, choose the best answer and blacken the corresponding space on the answer sheet.

Each question below consists of a word in capital letters, followed by five lettered words or phrases. Choose the word or phrase that is most nearly opposite in meaning to the word in capital letters. Since some of the questions require you to distinguish fine shades of meaning, consider all the choices before deciding which is best.

Example:

GOOD: (A) sour (B) bad (C) red (D) hot (E) ugly

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1. CHERISH: (A) despise (B) utilize (C) aspire (D) encourage (E) compete
2. VETO: (A) predict (B) discuss (C) approve (D) display (E) evaluate
3. EXTINGUISH: (A) graze (B) revive (C) correct (D) intrude (E) exceed
4. CEREMONIOUS: (A) active (B) enjoyable (C) permanent (D) informal (E) widespread
5. SYMMETRY: (A) exclusion (B) imbalance (C) isolation (D) immensity (E) validity
6. DOCUMENT: (A) edit (B) withhold (C) reproduce in full (D) write for pay (E) leave unsupported
7. HARBOR: (A) enlighten (B) burden (C) permit (D) prepare for (E) turn away
8. BREADTH: (A) rarity (B) mobility (C) complexity (D) narrowness (E) roughness
9. NOXIOUS: (A) diffuse (B) unique (C) beneficial (D) latent (E) static
10. REPREHENSIBLE: (A) matchless (B) praiseworthy (C) interesting (D) difficult to control (E) seldom recognized
11. SCANTY: (A) adept (B) copious (C) prosaic (D) candid (E) mellow
12. ADULATION: (A) initiation (B) vilification (C) injustice (D) purification (E) deliverance
13. PRODIGIOUS: (A) questionable (B) approximate (C) ultimate (D) adjacent (E) minuscule
14. TENSILE: (A) inelastic (B) genuine (C) tough (D) sympathetic (E) inharmonious
15. AMITY: (A) strife (B) irrelevance (C) realism (D) topicality (E) unseemliness

Each sentence below has one or two blanks, each blank indicating that something has been omitted. Beneath the sentence are five lettered words or sets of words. Choose the word or set of words that best fits the meaning of the sentence as a whole.

Example:

Although its publicity has been ----, the film itself is intelligent, well-acted, handsomely produced, and altogether ----.

(A) tasteless..respectable (B) extensive..moderate (C) sophisticated..amateur (D) risqué..crude (E) perfect..spectacular

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16. The ambassador's papers are not --- reading, but one who reads slowly and attentively will be richly repaid.

(A) petty (B) valuable (C) insightful (D) easy (E) plausible

17. It is inaccurate to describe Hopkins as a crusader for progressive reforms, for, although he debunks certain popular myths, he is not really --- of change.

(A) an advocate (B) a censor (C) an adversary (D) a caricature (E) a descendant

18. He was lonely and might have considered himself miserable were it not for a kind of hysterical ----, which he could neither account for nor ----.

(A) depression..enhance
(B) apathy..tolerate
(C) contentment..enjoy
(D) merriment..conquer
(E) sorrow..comprehend

19. Occasionally ---- strain of the bacteria appears, changed by some molecular misprint from what was once only ---- into a life-taking poison.

(A) a new..an epidemic
(B) a deficient..a derivative
(C) an erratic..a rudiment
(D) a virulent..a nuisance
(E) an advanced..a disease

20. The discussions were often ----, degenerating at times into name-calling contests.

(A) lofty (B) auspicious (C) acrimonious (D) lethargic (E) pragmatic

GO ON TO THE NEXT PAGE

Each passage below is followed by questions based on its content. Answer all questions following a passage on the basis of what is stated or implied in that passage.

I agree that children need to be—and usually want very much to be—taught right from wrong. But I believe that realistic fiction for children is one of the very hardest media in which it can be done. It is hard not to get entangled in simplistic moralism, and all you end up with are the "baddies" and the "goodies" you had hoped to avoid. Or you can trot out the cliché that "there's a little bit of bad in the best of us and a little bit of good in the worst of us," a dangerous trivialization of the fact that there is tremendous potential for good and for evil in every one of us. Or you can try the "problem books." The problem of drugs, of divorce, of prejudice, and so on—as if evil were merely a problem, something that can be solved, that has an answer, like a problem in fifth-grade arithmetic. That is escapism, that posing of evil as a "problem," instead of what it is: all the pain and suffering and injustice we will meet throughout our lives, and must admit, in order to live human lives at all.

But what, then, is the writer for children to do? Can one present the child with evil as an insoluble problem—something neither the child nor any adult can do anything about at all? To give the child a picture of a land haunted by famines or the cruelties of a brutal parent, and say, "Well, this is how it is. What are you going to make of it?"—that is surely unethical. If you suggest that there is a solution to these monstrous facts, you are lying to the child. If you insist that there is not, you are overwhelming the child with a burden that he or she is not strong enough yet to carry.

Children do need protection and shelter. But they also need the truth. And it seems to me that the way you can speak absolutely honestly and factually about both good and evil to children is to talk about themselves, their inner selves. That is something they can cope with; indeed, their job in growing up is to become themselves. They cannot do this if they feel the task is hopeless or if they are led to think there is no task.

Fantasy is the language of the inner self. I personally find it the appropriate language in which to tell stories to children—and others. I make that statement with some confidence, having behind me the authority of a very great poet, who put it much more boldly. "The great instrument of moral good," Shelley said, "is the imagination."

21. The author's primary goal in the passage is to do which of the following?
    (A) Criticize children's literature for being unrealistic
    (B) Demonstrate that morality is not a fit subject for children's literature
    (C) Argue that imaginative fiction is more suitable than realistic fiction for teaching morality to children
    (D) Propose a solution for the problem of evil in children's literature
    (E) Describe the way in which fantasy puts children in touch with the inner selves that they must obey

22. The author's attitude toward children appears to be one of
    (A) concern for the development of their moral integrity
    (B) idealization of their inexperience and vulnerability
    (C) contempt for their inability to accept unpleasant facts
    (D) exaggerated sympathy for their problems in daily life
    (E) envy of their willingness to learn about morality

23. According to the author, it is the duty of children's literature to
    (A) protect children from learning about anything unpleasant or distressing
    (B) simplify moral issues so that children can understand them
    (C) force children to deal with the facts of pain and suffering
    (D) reassure children that every problem has a solution
    (E) present truth in a way that children can accept and understand

24. In the passage the author indicates that much of human experience is
    (A) wretched and unbearable
    (B) difficult and unfair
    (C) routine and boring
    (D) unpredictable and meaningless
    (E) pleasant and secure

25. In presenting the argument, the author does all of the following EXCEPT
    (A) define a term
    (B) resolve a problem
    (C) refer to an authority
    (D) illustrate through an analogy
    (E) cite a psychological study

GO ON TO THE NEXT PAGE

One important facet of colonial America's early development was the role played by slave artisans from Africa. Slave art was an art of anonymity—an art that rose from slavery in the form of skilled handicrafts—an art descended from African imagery and later rechanneled as a functional aesthetic, finding its way to the wrought iron balconies of New Orleans and to the magnificent ornamentation on the mansions of Charleston, South Carolina. Although they received little credit for their labor, slave artisans accomplished necessary work in preindustrial America and became one of the first classes of technical experts in the New World.

Afro-American arts and crafts originated on the west coast of Africa where they were important in all aspects of communal life. Unlike the Europeans, who tended to perceive art objects only as "curios," Africans viewed the arts as functional and art objects were created for specific use in ceremonies as well as in numerous domestic activities. Many Africans were master artisans, demonstrating various skills and great proficiency in the fashioning of wood, bone, and ivory, in weaving, in pottery-making, and in the making of clothes, tools, and other implements. There is also strong evidence that some African groups were skilled in the building trades, whereas others proved skillful in developing exquisite sculptured objects of bronze through new casting techniques.

The major factors surrounding the development of the slave-artisan class included the diversification of farming and industry and the scarcity of labor to meet the needs of colonial America. As a solution to this problem, slaves were employed in every conceivable fashion. The diversity of occupations held by slave artisans is a clear example of why skilled slaves became very important agents in the rise of manufacturing.

Although research concerning slave artisans is lacking, certain general conclusions are apparent. Black artisans constituted a specialized labor force in colonial America. Without their achievements it is difficult to imagine how the colonists would have survived and prospered. As producers of goods, contributors to the building trades, manufacturers of furniture, and designers of household objects and decor, slave artisans also aided America in its aesthetic development. Their creativity and cleverness laid the foundation for furthering the development of the Afro-American artist.

26. The author is primarily concerned with

(A) describing the slave artisans' response to the limited opportunities for creative expression in colonial America
(B) discussing the slaves' artistic heritage and the role their skills played in preindustrial America
(C) depicting the specific talents of slave artisans in the areas of decoration and design
(D) noting that contemporary Black art emerged from the work of slave artisans
(E) praising the long history of the artistic tradition of Africa

27. It can be inferred that one reason slave art was "an art of anonymity" (line 3) was that the slave artisans

(A) received inadequate fees for their work
(B) were believed to do stylized work
(C) did not receive individual credit for their work
(D) were required to work only on specifically assigned projects
(E) relied on the artistic techniques of their ancestors

28. With which of the following statements regarding artisans in Africa would the author most likely agree?

(A) They were the most respected members of their societies.
(B) They were rewarded for their contributions to the community.
(C) Their skills related primarily to manufacturing activities.
(D) All members of a society were considered artisans.
(E) They played a vital role in communal life.

29. According to the passage, slave artisans were important to colonial America because they

(A) helped meet the needs of its preindustrial society
(B) persuaded the colonists to diversify farming and industry
(C) recognized the need for new and more complex manufacturing procedures
(D) taught other slaves skills that led to greater freedom
(E) helped to develop plantations into self-sufficient communities

30. The author cites specific examples of the work of slave artisans primarily to

(A) show that they had mastered basic craft skills
(B) indicate the conventional and imitative nature of their work
(C) explain why they were considered technical experts rather than artists
(D) attest to the quality and variety of their work
(E) emphasize the limited opportunities granted to them

GO ON TO THE NEXT PAGE

Select the word or set of words that best completes each of the following sentences.

31. Many sportswriters have been caught up in the activities about which they write and have become advocates and ---- when they ought to have been ----.

(A) promoters. .colleagues
(B) participants. .collaborators
(C) apologists. .critics
(D) opponents. .antagonists
(E) disputants. .defenders

32. His inability to fathom the latest trends in art led him to fear that his critical faculties had ---- during his long absence.

(A) diversified (B) atrophied (C) converted
(D) predominated (E) multiplied

33. Though her lecture contained ideas that were provocative and systematically presented, her style of delivery was so ---- that I actually dozed off.

(A) galling (B) pungent (C) desultory
(D) soporific (E) theatrical

34. Fuentes' subtly persuasive arguments for continuity in Latino culture ---- readers to recognize that their future cannot be ---- from the way they treat their past.

(A) implore. .deciphered
(B) condition. .inferred
(C) invite. .divorced
(D) command. .projected
(E) inspire. .elicited

35. Like most ---- literature, this moving remembrance of the poet's parents primarily expresses lamentation for their deaths.

(A) dogmatic (B) elegiac (C) abstract
(D) dramatic (E) striking

Each question below consists of a related pair of words or phrases, followed by five lettered pairs of words or phrases. Select the lettered pair that best expresses a relationship similar to that expressed in the original pair.

Example:

YAWN:BOREDOM :: (A) dream:sleep
(B) anger:madness (C) smile:amusement
(D) face:expression (E) impatience:rebellion

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36. JUDGE:COURTHOUSE :: (A) physician:hospital
(B) clergyman:library (C) farmer:house
(D) visitor:hotel (E) mathematician:computer

37. MUMBLE:INDISTINCT :: (A) relent:gentle
(B) stumble:graceful (C) enunciate:clear
(D) define:difficult (E) grunt:shrill

38. COLORS:SPECTRUM ::
(A) experiments:laboratory (B) panes:glass
(C) guests:party (D) letters:alphabet
(E) leaves:tree

39. TADPOLE:FROG :: (A) stream:river
(B) acorn:oak (C) politician:diplomat
(D) negative:photograph (E) student:graduate

40. ABODE:VAGRANT :: (A) ship:pirate
(B) fort:sentry (C) faith:prophet
(D) costume:eccentric (E) community:outcast

41. CREDITS:MOVIE :: (A) selections:album
(B) by-lines:newspaper (C) reviews:journal
(D) reruns:television (E) cartoons:government

42. AMORPHOUS:SHAPE :: (A) amorous:trust
(B) temporal:patience (C) enticing:guile
(D) bland:zest (E) classical:harmony

43. EVIL:MALEFACTOR ::

(A) selfishness:hermit
(B) talent:virtuoso
(C) benevolence:miser
(D) mischief:benefactor
(E) friendliness:thief

44. DIGRESS:SPEECH :: (A) dissemble:truth
(B) meander:travel (C) narrate:climax
(D) deter:progress (E) circumnavigate:globe

45. PLAINTIVE:SORROW ::
(A) systematic:brevity (B) elusive:eloquence
(C) confident:defeat (D) distasteful:pessimism
(E) contemptuous:disdain

IF YOU FINISH BEFORE TIME IS CALLED, YOU MAY CHECK YOUR WORK ON THIS SECTION ONLY. DO NOT WORK ON ANY OTHER SECTION IN THE TEST. **STOP**

SECTION 2 Time—30 minutes, 25 Questions

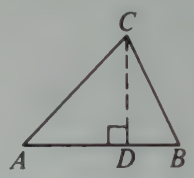
In this section solve each problem, using any available space on the page for scratchwork. Then decide which is the best of the choices given and blacken the corresponding space on the answer sheet.

The following information is for your reference in solving some of the problems.

Circle of radius r: Area $= \pi r^2$; Circumference $= 2\pi r$
The number of degrees of arc in a circle is 360.
The measure in degrees of a straight angle is 180.

Definitions of symbols:

$=$ is equal to	$\leqq$ is less than or equal to
$\neq$ is unequal to	$\geqq$ is greater than or equal to
$<$ is less than	$\parallel$ is parallel to
$>$ is greater than	$\perp$ is perpendicular to



Triangle: The sum of the measures in degrees of the angles of a triangle is 180.

If $\angle CDA$ is a right angle, then

(1) area of $\triangle ABC = \frac{AB \times CD}{2}$

(2) $AC^2 = AD^2 + DC^2$

Note: Figures that accompany problems in this test are intended to provide information useful in solving the problems. They are drawn as accurately as possible EXCEPT when it is stated in a specific problem that its figure is not drawn to scale. All figures lie in a plane unless otherwise indicated. All numbers used are real numbers.

1. If $x - 7 = 5 - x$, then $x =$

   (A) -6 (B) -1 (C) 1 (D) 6 (E) 12

2. A gymnast competed in a meet and received the following scores for three events: 9.5 for bars, 8.7 for balance beam, and 8.8 for floor routine. What is the average (arithmetic mean) of these three scores?

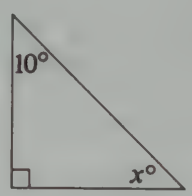
   (A) 8.9
   (B) 9.0
   (C) 9.1
   (D) 9.2
   (E) 9.3

3. On a number line, if point P has coordinate -3 and point Q has coordinate 5, what is the length of segment PQ?

   (A) 2 (B) 4 (C) 5 (D) 8 (E) 64

4. If $\frac{(20+50)+(30+N)}{2} = 70$, then $N =$

   (A) 30
   (B) 40
   (C) 50
   (D) 60
   (E) 70



Note: Figure not drawn to scale.

5. In the right triangle above, $x - 10 =$

   (A) 60
   (B) 70
   (C) 80
   (D) 90
   (E) 100

6. If $(x + 3)^2 = (x - 3)^2$, then $x =$

   (A) 0 (B) 1 (C) 3 (D) 6 (E) 9

GO ON TO THE NEXT PAGE

7. Ms. Jones borrowed \$1,000 for a year. The cost of the loan was 6 percent of the amount borrowed, to be paid back together with the loan at the end of the year. What was the total amount needed to pay off the loan?

(A) \$1,000.60
(B) \$1,006.00
(C) \$1,060.00
(D) \$1,600.00
(E) \$6,000.00

8. If $\frac{5}{x} = 1$ and $\frac{y}{2} = 3$, then $\frac{3+x}{y+3} =$

(A) $\frac{5}{6}$

(B) $\frac{8}{9}$

(C) 1

(D) $\frac{9}{8}$

(E) $\frac{6}{5}$

Candidate	Number of Votes Received
A	20
B	45
C	102
D	x
E	y

9. In a class of 300 students, 5 students were running for the position of student representative. If every student in the class voted for exactly one candidate and the distribution of votes is given in the table above, what is the maximum possible value of x ?

(A) 60 (B) 133 (C) 167
(D) 233 (E) 300

10. If $\lfloor x \rfloor$ is defined by the equation $\lfloor x \rfloor = \frac{\sqrt{x}}{2}$ for all whole numbers x, which of the following equals 5 ?

(A) $\lfloor 10 \rfloor$ (B) $\lfloor 20 \rfloor$ (C) $\lfloor 25 \rfloor$
(D) $\lfloor 50 \rfloor$ (E) $\lfloor 100 \rfloor$

11. To the nearest thousand, what is the number of seconds in a 24-hour day?

(A) 8,000
(B) 9,000
(C) 86,000
(D) 87,000
(E) 90,000



12. The figure above is a square divided into four equal smaller squares. If the perimeter of the large square is 1, then the perimeter of a small square is

(A) $\frac{1}{16}$ (B) $\frac{1}{8}$ (C) $\frac{1}{6}$ (D) $\frac{1}{4}$ (E) $\frac{1}{2}$

13. If $\frac{5}{6} = \frac{x}{5}$, then $x =$

(A) $\frac{6}{25}$ (B) $\frac{6}{5}$ (C) $\frac{25}{6}$ (D) 6 (E) 25

GO ON TO THE NEXT PAGE

14. A jar contains 10 pencils, some sharpened and some unsharpened. Each of the following could be the ratio of sharpened to unsharpened pencils EXCEPT

(A) 1 : 1
(B) 3 : 2
(C) 4 : 1
(D) 5 : 1
(E) 9 : 1

15. Cube X has a volume of 27. If point C is the center of one face of cube X and point D is the center of the opposite parallel face, what is the length of line segment CD?

(A) 3
(B) $3\sqrt{2}$
(C) 6
(D) $6\sqrt{2}$
(E) 9

16. Initially, there are exactly 18 bananas on a tree. If one monkey eats $\frac{1}{3}$ of the bananas and another monkey eats $\frac{1}{3}$ of the bananas that are left, how many bananas are still on the tree?

(A) 4
(B) 6
(C) 8
(D) 10
(E) 16

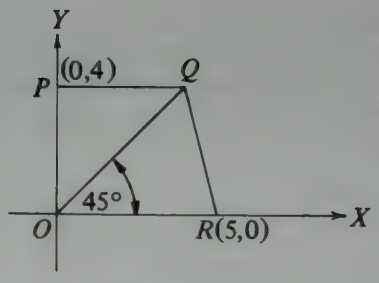
17. If $\frac{a}{b} + \frac{c}{d} = 4$, what is the value of $\frac{b}{a} + \frac{d}{c}$?

(A) $\frac{1}{4}$ (B) 1 (C) $\frac{4}{3}$ (D) 4

(E) It cannot be determined from the information given.

18. Which of the following is the greater of two numbers whose product is 220 and whose sum is 10 more than the difference between the two?

(A) 5 (B) 10 (C) 22 (D) 44 (E) 55



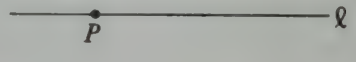
19. In the figure above, if $PQ \parallel OR$, what is the area of quadrilateral $PQRO$?

(A) 9 (B) 14 (C) 18 (D) 36

(E) It cannot be determined from the information given.

20. On the last day of a one-week sale, customers numbered 149 through 201 were waited on. How many customers were waited on that day?

(A) 51 (B) 52 (C) 53 (D) 152 (E) 153



21. If two points, Q and R, are each placed to the right of point P on line ℓ above so that $2PQ = 3PR$, what will be the value of $\frac{RQ}{PR}$?

(A) $\frac{1}{2}$ (B) $\frac{2}{5}$ (C) $\frac{2}{3}$ (D) $\frac{3}{2}$

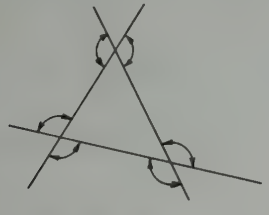
(E) It cannot be determined from the information given.

GO ON TO THE NEXT PAGE

X	Y
1	4
2	5
3	6

22. If x is a number from column X and y is a number from column Y in the table above, how many different values are possible for $x + y$?

(A) Nine (B) Eight (C) Seven
(D) Six (E) Five



23. Three lines intersect as shown in the figure above. What is the sum of the degree measures of the marked angles?

(A) 360° (B) 540° (C) 720° (D) 900°
(E) It cannot be determined from the information given.

24. If $x^7 = 3$ and $x^6 = \frac{2}{y}$, what is the value of x in terms of y ?

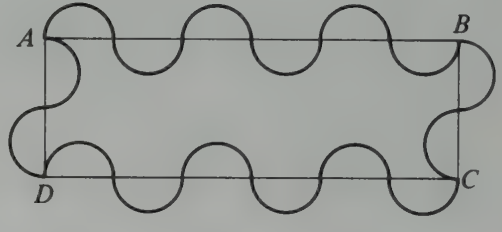
(A) $\frac{3}{2}y$

(B) $\frac{2}{3}y$

(C) $\frac{1}{3}y$

(D) $3y$

(E) $3 - \frac{2}{y}$



25. In the figure above, $ABCD$ is a rectangle and the curved path is made up of 16 semicircles of equal diameter. If the total length of this curved path is 32π, then the area of rectangle $ABCD$ is

(A) 24 (B) 32 (C) 48 (D) 64 (E) 192

IF YOU FINISH BEFORE TIME IS CALLED, YOU MAY CHECK YOUR WORK ON THIS SECTION ONLY. DO NOT WORK ON ANY OTHER SECTION IN THE TEST. **STOP**

SECTION 3 Time—30 minutes, 50 Questions

The questions in this section measure skills that are important to writing well. In particular, they test your ability to recognize and use language that is clear, effective, and correct according to the requirements of standard written English, the kind of English found in most college textbooks.

Directions: The following sentences contain problems in grammar, usage, diction (choice of words), and idiom.

Some sentences are correct.
No sentence contains more than one error.

You will find that the error, if there is one, is underlined and lettered. Assume that elements of the sentence that are not underlined are correct and cannot be changed. In choosing answers, follow the requirements of standard written English.

If there is an error, select the one underlined part that must be changed to make the sentence correct and blacken the corresponding space on your answer sheet.

If there is no error, blacken answer space (E).

EXAMPLE:

The region has a climate so severe that (A) plants growing there (B) rarely had been (C) more than twelve inches high (D). No error (E)

SAMPLE ANSWER

(A) (B) ● (D) (E)

1. A few scientists claim that food additives not only (A) improve the quality of (B) foods but also made (C) them (D) safer. No error (E)

2. It is rumored that (A) the names of them (B) to be promoted (C) will be announced tomorrow, but I believe the choices have not been made (D) yet. No error (E)

3. In 1957, the appearance of the first (A) Soviet satellite has created (B) a panic in the United States that continued for (C) nearly (D) a decade. No error (E)

4. In appreciation about (A) her work, the committee presented its (B) retiring director with, among other (C) gifts, a plaque describing (D) her accomplishments. No error (E)

5. For many people (A), hang-gliding, an increasingly (B) popular sport, seems satisfying (C) the urge to fly (D). No error (E)

6. Despite (A) much research, there are (B) still certain elements in (C) the life cycle of the cicada that are not fully understood (D). No error (E)

7. Louise's fair skin was (A) sunburned so badly (B) that she looked as if (C) she had fell into (D) a bucket of red paint. No error (E)

8. Whenever (A) we hear of a natural disaster, even in (B) a distant part of the world, you (C) feel sympathy (D) for the people affected. No error (E)

GO ON TO THE NEXT PAGE

9. Late in the war, the Germans, retreating (A) in haste (B), left many (C) of their (D) prisoners go free. No error (E)

10. In many states (A), there seems (B) to be a belief, openly (C) expressed by educators, that the methods of teaching reading should be changed (D). No error (E)

11. Throughout (A) the Middle Ages women work (B) beside (C) men, knowing that the efforts of men and women alike were essential to (D) survival. No error (E)

12. Without hardly a doubt (A), the novels of Thomas Pynchon are (B) more complex (C) than the novels of many other (D) contemporary writers. No error (E)

13. The leading roles in the widely acclaimed (A) play, a modern (B) version of an Irish folktale, were performed (C) by Jessica and he (D). No error (E)

14. In the reserve section of the library, there is (A) two volumes of (B) essays by James Baldwin on (C) the relation of the Black artist and intellectual to (D) society. No error (E)

15. Many studies have tried to (A) determine whether or not (B) seeing violence on television makes (C) children behave more violent (D). No error (E)

16. Between ten and (A) twenty per cent of the textbook appears to be (B) new; the rest (C) is a revision of (D) the previous edition. No error (E)

17. Some people seem remarkably (A) insensitive to physical pain, and (B) this insensitivity does not mean (C) that they are able to (D) endure other kinds of pain. No error (E)

18. During the meeting (A), Congresswoman Barbara Jordan stressed that (B) educators and legislators must cooperate where (C) the goal of equal opportunity is (D) to be reached. No error (E)

19. The population of American alligators, dangerously (A) small a few years ago, are (B) now estimated at (C) more than (D) one million. No error (E)

20. Such (A) novels as *Heidi* and *Little Women* have long been considered (B) by young and old alike (C) to be a classic of (D) children's literature. No error (E)

21. Anyone who (A) gathers mushrooms for the purpose of eating them (B) must distinguish carefully (C) between (D) poisonous and nonpoisonous species. No error (E)

22. In regards to (A) the energy crisis, the President urged (B) all homeowners to keep (C) their (D) thermostats at sixty-five degrees in winter. No error (E)

23. Long ago, well before (A) the invention of (B) the printing press, poets often sung (C) their (D) poetry to small, interested audiences. No error (E)

24. Because of (A) extreme weather conditions, starvation exists (B) in some countries where they (C) must struggle every day to stay alive (D). No error (E)

25. The energy question, along with several other (A) issues, are going (B) to be discussed (C) at (D) the next meeting of the state legislature. No error (E)

GO ON TO THE NEXT PAGE

Directions: In each of the following sentences, some part or all of the sentence is underlined. Below each sentence you will find five ways of phrasing the underlined part. Select the answer that produces the most effective sentence, one that is clear and exact, without awkwardness or ambiguity, and blacken the corresponding space on your answer sheet. In choosing answers, follow the requirements of standard written English. Choose the answer that best expresses the meaning of the original sentence.

Answer (A) is always the same as the underlined part. Choose answer (A) if you think the original sentence needs no revision.

EXAMPLE:

Laura Ingalls Wilder published her first book and she was sixty-five years old then.

(A) and she was sixty-five years old then
(B) when she was sixty-five years old
(C) at age sixty-five years old
(D) upon reaching sixty-five years
(E) at the time when she was sixty-five

SAMPLE ANSWER

Ⓐ ● Ⓒ Ⓓ Ⓔ

26. Freud's complex theory based on the death instinct is, for most people, one that is with difficult understanding.

(A) with difficult understanding
(B) difficult to understand
(C) to understand difficultly
(D) having difficulty being understood
(E) difficult for understanding

27. In the nineteenth century, trains were more than machines they were expressions of the greatness of the United States.

(A) machines they were
(B) machines and were
(C) machines; they were
(D) machines, although they were
(E) machines, but were

28. In the sunlight, the cherry blossoms that burst out everywhere, like foam on breaking waves.

(A) blossoms that burst
(B) blossoms bursting
(C) blossoms, which are bursting
(D) blossoms burst
(E) blossoms, which burst

29. When Dorothy Richardson decided to become a novelist, she knew that her writing would leave her little time for other work.

(A) When Dorothy Richardson decided to become
(B) After the decision was made by Dorothy Richardson to become
(C) After the decision by Dorothy Richardson to become
(D) When Dorothy Richardson decides to become
(E) When Dorothy Richardson decided about becoming

30. The difference between the twins is that one is humorous; the other, serious.

(A) one is humorous; the other, serious
(B) of one being humorous, the other is serious
(C) one is humorous; the other being serious
(D) one is humorous, although the other is more serious
(E) of a humorous one and one that is serious

31. The United States did not go on the gold standard until 1900, it went off it thirty-three years later.

(A) until 1900, it went off it thirty-three years later
(B) until 1900; however, going off after thirty-three years
(C) until 1900, although going off in thirty-three years
(D) until 1900, it was thirty-three years later when it went off it
(E) until 1900 and went off it thirty-three years later

32. Many memos were issued by the director of the agency that had an insulting tone, according to the staff members.

(A) Many memos were issued by the director of the agency that
(B) Many memos were issued by the director of the agency who
(C) The issuance of many memos by the director of the agency which
(D) The director of the agency issued many memos that
(E) The director of the agency, who issued many memos that

GO ON TO THE NEXT PAGE

33. The widespread slaughter of buffalo for their skins profoundly shocked and angered Native Americans.

(A) shocked and angered Native Americans
(B) shocked Native Americans, angering them
(C) shocked Native Americans, and they were also angry
(D) was a shock and caused anger among Native Americans
(E) was shocking to Native Americans, making them angry

34. Consumers are beginning to take notice of electric cars because they are quiet, cause no air pollution, and gasoline is not used.

(A) cause no air pollution, and gasoline is not used
(B) air pollution is not caused, and gasoline is not used
(C) cause no air pollution, and use no gasoline
(D) causing no air pollution and using no gasoline
(E) air pollution is not caused, and no gasoline is used

35. Because she was a woman was why Sharon Frontiero, a lieutenant in the United States Air Force, felt that she was being treated unfairly.

(A) Because she was a woman was why Sharon Frontiero, a lieutenant in the United States Air Force, felt that she was being treated unfairly.
(B) Sharon Frontiero, a lieutenant in the United States Air Force, felt that she was being treated unfairly because she was a woman.
(C) Because she was a woman, Sharon Frontiero felt that this was why she was being treated unfairly as a lieutenant in the United States Air Force.
(D) Sharon Frontiero, a lieutenant in the United States Air Force, feeling that she was being treated unfairly because she was a woman.
(E) A woman, Sharon Frontiero, felt that because she was a lieutenant in the United States Air Force, that she was being treated unfairly.

36. A major difference between people and apes is brain size, a person's brain is three times as large as the brain of an ape.

(A) size, a person's brain is
(B) size, with a person's brain being
(C) size; a person's brain is
(D) size; a person's brain, it is
(E) size, in that it is

37. The process of how European immigrant groups being absorbed into American society is complex.

(A) of how European immigrant groups being absorbed
(B) for European immigrant groups being absorbed
(C) where European immigrant groups were absorbed
(D) by which European immigrant groups have been absorbed
(E) whereby the absorption of European groups has been

38. Mr. Howe's class has organized a special program for our school; the purpose being to help us increase our understanding of Japanese culture.

(A) school; the purpose being to
(B) school and the purpose is to
(C) school, the purpose is to
(D) school, being to
(E) school to

39. Light reaching earth from the most distant stars originated billions of years ago.

(A) reaching earth from the most distant stars
(B) which reaching earth from the most distant stars
(C) from the most distant stars reaching earth
(D) that is from the most distant stars and reaches earth
(E) reaching earth which is from stars that are most distant

40. Joseph Conrad was born and educated in Poland and he wrote all of his novels in English.

(A) Joseph Conrad was born and educated in Poland and he
(B) Joseph Conrad, being born and educated in Poland,
(C) Although being born and educated in Poland, Joseph Conrad
(D) Although Joseph Conrad was born and educated in Poland, he
(E) Being from Poland, where he was born and educated, Joseph Conrad

GO ON TO THE NEXT PAGE

Note: The remaining questions are like those at the beginning of the section.

Directions: For each sentence in which you find an error, select the one underlined part that must be changed to make the sentence correct and blacken the corresponding space on your answer sheet.

If there is no error, blacken answer space (E).

EXAMPLE:

The region has a climate so severe that (A) plants growing there (B) rarely had been (C) more than twelve inches high (D). No error (E)

SAMPLE ANSWER

(A) (B) ● (D) (E)

41. If you find that it is (A) (B) difficult to concentrate in noisy surroundings (C), one should (D) try to find a quiet place to study. No error (E)

42. The speaker claimed that no other modern nation devotes (A) so small a portion of (B) its wealth (C) to public assistance and health as (D) the United States does. No error (E)

43. The thought of trying to (A) persuade their (B) three-year-old to sit in (C) a high chair did not appeal to either the mother nor (D) the father. No error (E)

44. The bright (A) fiberglass sculptures of Luis Jiménez has received (B) critical acclaim not only (C) in his home state, New Mexico, but also in (D) New York. No error (E)

45. Doctors see a connection between (A) increased (B) amounts of leisure time spent sunbathing and (C) the increased number of cases of skin cancer (D). No error (E)

46. Whether or not (A) credit card companies should prevent their (B) customers to acquire (C) substantial debts was the issue discussed at (D) the meeting. No error (E)

47. The board's final recommendations included (A) hiring additional (B) personnel, dismissing the head of (C) research, and a reorganized marketing (D) division. No error (E)

48. Some people prefer (A) attending movies to television (B) because they (C) dislike the frequent (D) interruptions of programs for commercials. No error (E)

49. Like (A) many factory workers of (B) a century ago, women today (C) are developing organizations to represent their interests (D). No error (E)

50. In a prominent city newspaper, they claim (A) that the number of (B) unregistered participants in (C) this year's marathon was half that of (D) last year's. No error (E)

IF YOU FINISH BEFORE TIME IS CALLED, YOU MAY CHECK YOUR WORK ON THIS SECTION ONLY. DO NOT WORK ON ANY OTHER SECTION IN THE TEST. **STOP**

SECTION 4 Time—30 minutes, 40 Questions. For each question in this section, choose the best answer and blacken the corresponding space on the answer sheet.

Each question below consists of a word in capital letters, followed by five lettered words or phrases. Choose the word or phrase that is most nearly opposite in meaning to the word in capital letters. Since some of the questions require you to distinguish fine shades of meaning, consider all the choices before deciding which is best.

Example:

GOOD: (A) sour (B) bad (C) red (D) hot (E) ugly

Ⓐ ● Ⓒ Ⓓ Ⓔ

1. RECALL: (A) oppose (B) forget (C) injure (D) assist (E) quiet

2. INTERCEDE: (A) render harmless (B) stand aside (C) direct incompetently (D) protect publicly (E) convene hastily

3. GRANDIOSE: (A) attractive (B) unhealthy (C) bright and shiny (D) small and unimpressive (E) soft and manageable

4. UNNERVE: (A) warn (B) release (C) evaluate (D) strengthen (E) believe

5. RUFFIANISM: (A) gentle behavior (B) cold disdain (C) false piety (D) ignorant statement (E) wishful thinking

6. PALPABLE: (A) retroactive (B) decorative (C) imperturbable (D) not tangible (E) not trustworthy

7. MIRED: (A) dismissed (B) corrected (C) unsound (D) unhampered (E) unregulated

8. DISPARAGE: (A) praise profusely (B) surrender silently (C) fraternize (D) exorcise (E) reprieve

9. DISSONANCE: (A) practicality (B) agreement (C) probability (D) loyalty (E) cheerfulness

10. CAJOLE: (A) animate (B) browbeat (C) measure up to (D) work intensively (E) determine rapidly

Each sentence below has one or two blanks, each blank indicating that something has been omitted. Beneath the sentence are five lettered words or sets of words. Choose the word or set of words that best fits the meaning of the sentence as a whole.

Example:

Although its publicity has been ----, the film itself is intelligent, well-acted, handsomely produced, and altogether ----.

(A) tasteless..respectable (B) extensive..moderate (C) sophisticated..amateur (D) risqué..crude (E) perfect..spectacular

● Ⓑ Ⓒ Ⓓ Ⓔ

11. Medieval kingdoms did not become constitutional republics overnight; on the contrary, the change was ----.

(A) unpopular (B) unexpected (C) advantageous (D) sufficient (E) gradual

12. This carefully researched book is useful because it ---- the ---- concerns and influences that have connected women writers over three centuries.

(A) omits. .essential
(B) distorts. .ephemeral
(C) charts. .underlying
(D) dispels. .unimpeachable
(E) foresees. .documented

13. The reef's fragile surface of living polyps is probably more ---- to wounds and infection than a child's skin; indeed, merely brushing against living coral ---- its delicate protoplasm.

(A) resistant. .revives (B) susceptible. .enhances (C) immune. .imperils (D) vulnerable. .damages (E) attractive. .impairs

GO ON TO THE NEXT PAGE

14. Though discrepant popular belief makes the findings appear to be ----, recent behavioral studies suggest that most teen-agers are actually happy.

(A) discriminating (B) heretical
(C) anticipated (D) obligatory
(E) depressing

15. The validity of her experimental findings was so ---- that even the most ---- investigators could not refrain from extolling her.

(A) enigmatic. .officious
(B) fallacious. .credulous
(C) unassailable. .disputatious
(D) inevitable. .convivial
(E) dubious. .skeptical

Each question below consists of a related pair of words or phrases, followed by five lettered pairs of words or phrases. Select the lettered pair that best expresses a relationship similar to that expressed in the original pair.

Example:

YAWN:BOREDOM :: (A) dream:sleep
(B) anger:madness (C) smile:amusement
(D) face:expression (E) impatience:rebellion

Ⓐ Ⓑ ● Ⓓ Ⓔ

16. HEIGHT:MOUNTAIN :: (A) depth:trench
(B) shade:tree (C) weight:age
(D) speed:highway (E) mineral:mine

17. CENSUS:POPULATION ::
(A) election:government (B) criterion:judgment
(C) inventory:stock (D) drought:thirst
(E) recipe:cake

18. POET:WORDS :: (A) sculptor:stone
(B) painter:artistry (C) sailor:ocean
(D) physician:care (E) philosopher:book

19. CONTEMPLATION:THINKER ::
(A) corrosion:chemist (B) construction:builder
(C) sullenness:fighter (D) hesitation:liar
(E) pain:soldier

20. RIDDLED:HOLES :: (A) untangled:knots
(B) wrinkled:materials (C) flawed:repairs
(D) pitted:indentations (E) sharpened:injuries

21. MUSE:INSPIRATION :: (A) editor:personality
(B) model:criticism (C) epic:superstition
(D) plot:characterization (E) patron:support

22. VEHEMENT:ENERGY ::
(A) domineering:caution
(B) compassionate:sympathy
(C) dauntless:fear
(D) ruthless:suspicion
(E) amiable:apathy

23. FLAGRANT:OBSERVER ::
(A) blaring:listener
(B) monotonous:speaker
(C) mischievous:prankster
(D) temporary:visitor
(E) aggressive:attacker

24. ASTUTE:INSIGHT :: (A) dutiful:efficiency
(B) affable:friendliness (C) gullible:dependence
(D) zealous:nobility (E) proud:ambition

25. LABOR:RESPITE :: (A) opinion:dissent
(B) commitment:betrayal (C) action:resentment
(D) error:erasure (E) debate:lull

GO ON TO THE NEXT PAGE

Each passage below is followed by questions based on its content. Answer all questions following a passage on the basis of what is stated or implied in that passage.

However one chooses to define weather and climate, there is a simple practical distinction between them: weather is a matter of everyday experience, whereas climate is a statistical generalization of that experience. Weather is what usually interests us, but to interpret it correctly we must first determine what the underlying climate is in general; otherwise, we have no yardstick for comparison. In a sense, climate is like the carrier wave in radio broadcasting on which the irregular signals of programming–weather–are superimposed. We have to study the "normal" distribution of the meteorological elements before we can begin to understand the "variability around the norm" that constitutes weather.

Of all the climatic elements, solar radiation is the most basic. The intensity and duration of sunshine is a partial indication of this radiation, but there are also invisible parts of the radiation that have not by any means been as closely studied. Cloud cover is possibly the next most important because it determines both how much solar radiation reaches the ground and how much is radiated back into space from the Earth's surface. Temperature, which depends on radiation, influences cloud cover and controls humidity, and is obviously important in itself. Winds and pressure are interdependent, the latter representing the weight of the atmosphere, the former representing the air movement that ensues when adjacent vertical columns of the atmosphere have different weights. Humidity indicates the amount of water in vapor form in the atmosphere; precipitation measures the amount of water in its liquid or solid phase that falls to the Earth's surface.

26. Which of the following is the best title for the passage?

(A) Weather and Its Consequences
(B) Technology and Climate
(C) Can Weather Be Controlled?
(D) Climate: A Mixed Blessing
(E) The Components of Climate

27. Which of the following comparisons most closely parallels the relationship between weather and climate described in the first paragraph?

(A) A particular game to a team's season record
(B) A heat-seeking missile to a fighter plane
(C) A bacterium to its colony
(D) An element to a compound
(E) A waterfall to a lake

28. It can be inferred from the passage that the definition of precipitation includes which of the following?

I. Water vapor
II. Rain and snow
III. Hail and sleet

(A) II only (B) I and II only
(C) I and III only (D) II and III only
(E) I, II, and III

GO ON TO THE NEXT PAGE

"It seems very odd, very sad," Margaret returned, "that you can never act unselfishly in society affairs. If I wished to go and see those girls just to do them a pleasure, and perhaps because they're new in town and lonely, I might do them good, even—but it would be impossible."

"Quite," said her aunt. "Such a thing would be quixotic. Society doesn't rest upon any such basis. It can't; it would go to pieces if people acted from unselfish motives."

"Then it isn't society at all!" said the girl. "All its favors are really bargains. Its gifts are for gifts back again."

"Yes, that is true," said Mrs. Horn, with no more sense of a judgment in the fact than the political economist has in the fact that wages are the measure of necessity and not of merit. "You get what you pay for. It's a matter of business." She satisfied herself with this formula, but she did not dislike her niece's revolt against it. That was part of Margaret's originality, which pleased her aunt in proportion to her own conventionality; she was really a timid person, and she liked the show of courage which Margaret's magnanimity often reflected upon her. She thought that she set bounds to the girl's originality because she recognized them. Margaret understood this better than her aunt and knew that she had consulted her about going to see the girls out of deference and with no expectation of luminous instruction.

29. Mrs. Horn would most likely consider which of the following a good example of the way society actually works?

(A) A celebrity lavishly supports a charity organization solely to obtain favorable publicity.
(B) A city government intervenes to halt construction of a shopping center in a residential district.
(C) A parent provides a child with high ideals but cannot translate theory into practice.
(D) A building contractor secretly replaces expensive materials with shabby and inadequate goods.
(E) A doctor receives a prestigious award for the discovery of a cure for a dangerous disease.

30. According to the passage, Margaret asked Mrs. Horn's opinion because she

(A) wished to emulate her aunt's behavior in society
(B) could not choose a suitable course of action
(C) thought she would receive good advice
(D) needed to bolster her aunt's self-confidence
(E) wanted to show respect for her aunt

31. According to the passage, Mrs. Horn sees her relationship with Margaret as one in which she

(A) protects Margaret from the natural evil of society
(B) consoles Margaret in her disillusionment and loss of innocence
(C) enlightens Margaret by pointing out limitations of her ideas
(D) humiliates Margaret through constant proof that she is wrong
(E) irritates Margaret by frequent intrusions into her personal life

GO ON TO THE NEXT PAGE

Rodents are the largest order of mammals, both in number of species and in individuals, outnumbering all other warm-blooded quadrupeds and bipeds combined. There is considerable variation in this order, which includes beavers, porcupines, rats, lemmings, badgers, and the familiar cage pets, gerbils and hamsters. Rodents are ubiquitous, inhabiting almost every continent, either as natives or as immigrants. Most are hypertense and forever on the alert for predators, since rodents are the principal item on the menus of most furred and feathered carnivores. The common denominator among rodents is their teeth—oversize, chisel-like incisors that grow constantly throughout their lives and that are kept sharp and trimmed by constant chewing on wood, nuts, plaster walls, or other hard matter. Untrimmed, their teeth will grow into inward-curving tusks that can seriously injure or even kill them.

The most familiar rodents are rats and mice, whose usefulness as laboratory animals and as important links in the food chain is virtually unrivaled. In laboratories all over the world, domesticated rats and mice are the prime subjects of scientific experiments and research projects that aim to cure human diseases and determine the effects of myriad drugs on human beings. Because rodents are similar to humans in the adaptability of their eating habits, rats and mice are invaluable to scientists doing studies on diet and nutrition. Another asset is the rodents' life span; in the wild, rats and mice rarely live longer than one year because they are preyed on by a large number of animals, including snakes, dogs, cats, owls, and hawks, but in captivity they may live as long as three years. Such a period is just right for studies on aging, growth, and heredity. In addition, rats and mice are a perfect size to house and handle with ease in laboratories. In the United States alone, scientific studies use some 18 million rats each year. It is the rare person who has not reaped, directly or indirectly, the benefits of the medical and psychological research done with these adaptable rodents.

To most people, the destructiveness and havoc rats and mice wreak on the environment far outweigh any of their virtues. Yet the vast majority of rodents are actually beneficial and essential to the overall balance of nature; they furnish food for other animals and prey on insects, whose numbers they keep in check. These rodents should never be confused with the true villains of the order: the brown or Norway rat, the black rat, and the innocent-looking house mouse. As carriers of plague, typhus, and other epidemic diseases, this trio has inflicted death and misery on the world since prehistoric times. The Black Death, the most catastrophic plague in history, killed approximately one-quarter of the population of medieval Europe and was almost certainly spread by rats, which serve as hosts for plague-bearing fleas.

32. The author states that the laboratory rodents' life span is especially helpful to scientists studying

(A) the effect of captivity on laboratory animals
(B) means of controlling the rodent population
(C) population distribution and predatory patterns
(D) genetic characteristics and growth patterns
(E) the social habits of rodents

33. The passage mentions all of the following facts about rodents EXCEPT that they

(A) eat insects
(B) are amphibious
(C) are easy to handle
(D) chew constantly
(E) can eat a variety of foods

34. It can be inferred that the author thinks rodents are useful for all of the following reasons EXCEPT that they

(A) tend to be very alert
(B) can make excellent laboratory animals
(C) are similar to humans in some ways
(D) are a source of food for other animals
(E) are an important part of the balance of nature

35. According to the author, which of the following is (are) true of rodents?

I. They are a staple in the diets of many predators.
II. Many laboratory rodents live longer than wild rodents do.
III. Most species spread devastating diseases.

(A) I only
(B) II only
(C) III only
(D) I and II only
(E) II and III only

36. Which of the following best describes the development of this passage?

(A) Major points, minor points
(B) Statement of problem, examples, proposed solution
(C) Introduction, positive factors, negative factors
(D) Introduction, cause, results
(E) Thesis, analogy, antithesis

GO ON TO THE NEXT PAGE

"What if all scientists were to publish anonymously?" I was asked this question recently by Dr. Lester Green. "Don't you think," he continued, "that scientific literature would be far less cluttered with nearly useless and carelessly produced articles were the authors to receive no public credit for their contributions? We might, in fact, be able to achieve that great nirvana in which science would be practiced for its own sake, rather than for fame and fortune."

Line
(5)
(10)

What lies at the heart of the notion pondered by Green is the idea that individual and community interests are necessarily separate entities. Does Green seriously believe that the best interests of society would be served by depersonalizing science? Would he suggest that artists not sign their paintings?

(15)

"The highest form of vanity is love of fame," wrote George Santayana. But is an ambition for fame the worst reason for practicing science? Dr. Samuel Johnson said, "None but a blockhead ever wrote except for money." We scientists might paraphrase him, "None but a blockhead ever published a scientific article except for recognition."

(20)

Furthermore, anonymity leads to secrecy and secrecy in science is deplorable. Even when countries are at war, recognition of individual performance is necessary. True, scientists have, during periods of crisis such as the Second World War, sacrificed public recognition. In time, however, most of the important breakthroughs were credited to their discoverers.

(25)
(30)

37. The author's purpose in the passage is apparently to

(A) trace the history of an idea
(B) merge two differing views of an issue
(C) discredit the majority of research scientists
(D) argue against a proposed change
(E) demand a new set of standards

38. It can be inferred that Dr. Green is dissatisfied with the

(A) desire for anonymity among noted scientists
(B) overall quality of articles appearing in scientific literature
(C) small number of articles published by scientists
(D) recent emphasis on secrecy in the scientific community
(E) lack of recognition given to scientists who publish articles

39. Which of the following best describes the author's attitude toward the "nirvana" (line 8) mentioned by Dr. Green?

(A) Excited enthusiasm
(B) Indulgent tolerance
(C) Fascinated curiosity
(D) Cautious skepticism
(E) Disapproving dismissal

40. The author implies that an artist who did not sign his or her painting should be regarded as

(A) a fool (B) a martyr (C) an egotist
(D) a conformist (E) a pioneer

IF YOU FINISH BEFORE TIME IS CALLED, YOU MAY CHECK YOUR WORK ON THIS SECTION ONLY. DO NOT WORK ON ANY OTHER SECTION IN THE TEST. **STOP**

SECTION 5 Time—30 minutes, 35 Questions

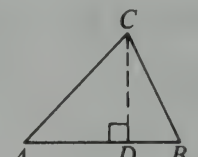
In this section solve each problem, using any available space on the page for scratchwork. Then decide which is the best of the choices given and blacken the corresponding space on the answer sheet.

The following information is for your reference in solving some of the problems.

Circle of radius r: Area $= \pi r^2$; Circumference $= 2\pi r$
The number of degrees of arc in a circle is 360.
The measure in degrees of a straight angle is 180.

Definitions of symbols:

$=$ is equal to	$\leqq$ is less than or equal to
$\neq$ is unequal to	$\geqq$ is greater than or equal to
$<$ is less than	$\parallel$ is parallel to
$>$ is greater than	$\perp$ is perpendicular to



Triangle: The sum of the measures in degrees of the angles of a triangle is 180.

If $\angle CDA$ is a right angle, then

(1) area of $\triangle ABC = \frac{AB \times CD}{2}$

(2) $AC^2 = AD^2 + DC^2$

Note: Figures that accompany problems in this test are intended to provide information useful in solving the problems. They are drawn as accurately as possible EXCEPT when it is stated in a specific problem that its figure is not drawn to scale. All figures lie in a plane unless otherwise indicated. All numbers used are real numbers.

1. A man's grocery bill is \$8, but the store deducts \$1.50 from his bill for coupons. If the man gives the grocery clerk \$10, how much change should he get?

   (A) \$8.50
   (B) \$6.50
   (C) \$5.50
   (D) \$4.50
   (E) \$3.50

2. Which of the following numbers is divisible by 3 and 5, but not by 2 ?

   (A) 955
   (B) 975
   (C) 990
   (D) 995
   (E) 999

3. The members of a club decided to wash cars in order to earn money for the club. Each member of the club washed 3 cars and charged \$2 per car. When they had finished, their receipts totalled \$66, which included \$6 in tips. How many members were in the club?

   (A) 9
   (B) 10
   (C) 11
   (D) 20
   (E) 22

4. If x and y are positive integers such that $x^2 + y^2 = 13$ and $x > y$, then $x - y =$

   (A) 1 (B) 2 (C) 3 (D) 4 (E) 5

5. If the perimeter of a rectangular garden plot is 80 meters, which of the following could be the length of one of its sides?

   I. 30 meters
   II. 40 meters
   III. 50 meters

   (A) I only (B) II only (C) III only
   (D) I and II (E) II and III

6. A wire of uniform diameter and composition that weighs 32 pounds is cut into two pieces. One piece is 90 yards long and weighs 24 pounds. What was the length, in yards, of the original wire?

   (A) 60
   (B) 120
   (C) 135
   (D) 270
   (E) 360

7. If $\frac{1}{2}$ of a number is 2 more than $\frac{1}{3}$ of the number, what is the number?

   (A) 2
   (B) 6
   (C) 12
   (D) 20
   (E) 24

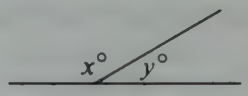
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Questions 8-27 each consist of two quantities, one in Column A and one in Column B. You are to compare the two quantities and on the answer sheet blacken space

A if the quantity in Column A is greater;
B if the quantity in Column B is greater;
C if the two quantities are equal;
D if the relationship cannot be determined from the information given.

AN E RESPONSE WILL NOT BE SCORED.

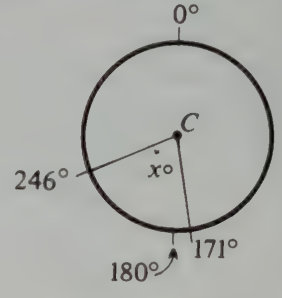
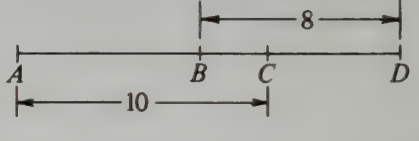
EXAMPLES

	Column A	Column B	Answers
E1.	2×6	$2 + 6$	● Ⓑ Ⓒ Ⓓ Ⓔ
E2.	 $180 - x$	y	Ⓐ Ⓑ ● Ⓓ Ⓔ
E3.	$p - q$	$q - p$	Ⓐ Ⓑ Ⓒ ● Ⓔ

Notes:

1. In certain questions, information concerning one or both of the quantities to be compared is centered above the two columns.
2. In a given question, a symbol that appears in both columns represents the same thing in Column A as it does in Column B.
3. Letters such as x, n, and k stand for real numbers.

	Column A	Column B
8.	$\frac{1}{3} - \frac{1}{5}$	$\frac{2}{15}$
	n is a negative integer.	
9.	$n \times n \times n \times n$	$n + n + n + n$
	$P = \{1, 2, 3, 4\}$ $Q = \{1, 2, 3, 4, 5\}$	
10.	The number that is a member of set Q but not of set P	A number that is a member of both sets P and Q
	$x = \frac{1}{4}$ $y = \frac{1}{2}$	
11.	$\frac{x}{y}$	$\frac{y}{x}$
12.	The number of edges of a cube	The number of faces of a cube
	n is an integer.	
13.	The remainder when n is divided by 9	The remainder when n is divided by 6

	Column A	Column B
14.	Average (arithmetic mean) of −9, −8, 8, and 9	Average (arithmetic mean) of −7, −6, 0, 6, and 7
	 C is the center of the circle. $x > 0$	
15.	x	66
		
16.	Length of AD	18

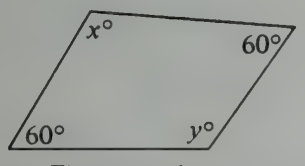
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SUMMARY DIRECTIONS FOR COMPARISON QUESTIONS

Answer: A if the quantity in Column A is greater;
B if the quantity in Column B is greater;
C if the two quantities are equal;
D if the relationship cannot be determined from the information given.

AN E RESPONSE WILL NOT BE SCORED.

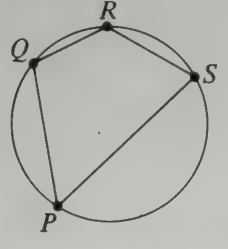
	Column A	Column B
	$-5 < x < -3$ $-7 < y < -5$ x and y are even integers.	
17.	x	y

	Column A	Column B
	 Note: Figure not drawn to scale.	
18.	x	y

	Column A	Column B
	$x - 2 = y$	
19.	$x - 4$	y

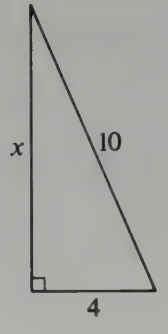
	Column A	Column B
20.	$\frac{4}{\sqrt{2}}$	$2\sqrt{2}$

	Column A	Column B
	$x = 501$ $y = 500$	
21.	$(x + y)(x - y)$	1,000

	Column A	Column B
	 The diameter of the circle is 1. P, Q, R, and S are on the circumference of the circle.	
22.	Perimeter of quadrilateral $PQRS$	4

	Column A	Column B
	A fair six-sided die with faces numbered 1 through 6 is to be rolled twice.	
23.	The probability of obtaining a 6 on the top face on the first roll and a 5 on the top face on the second roll	The probability of obtaining a 5 on the top face on both the first and second rolls

	Column A	Column B
	x and y are consecutive terms of a sequence whose terms alternate, 1, −2, 1, −2, 1, −2, etc.	
24.	$x + y$	$x - y$

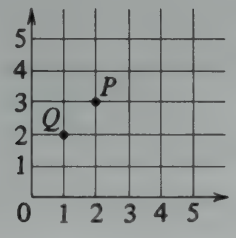
	Column A	Column B
		
25.	x	9

	Column A	Column B
26.	$0.7x + 0.9y$	$0.7(x + y)$

	Column A	Column B
	a, b, and c are positive.	
27.	Average (arithmetic mean) of a, b, and c	Average (arithmetic mean) of a^2, b^2, and c^2

GO ON TO THE NEXT PAGE

Solve each of the remaining problems in this section using any available space for scratchwork. Then decide which is the best of the choices given and blacken the corresponding space on the answer sheet.

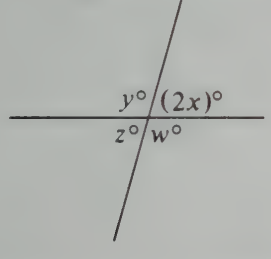


28. Which of the following points, when plotted on the grid above, will be twice as far from $P(2, 3)$ as from $Q(1, 2)$?

(A) $(0, 1)$ (B) $(0, 3)$ (C) $(1, 3)$
(D) $(3, 4)$ (E) $(3, 5)$

29. If 15 kilograms of pure water is added to 10 kilograms of pure alcohol, what percent by weight of the resulting solution is alcohol?

(A) $66\frac{2}{3}\%$ (B) 40% (C) 25%
(D) 15% (E) 10%



30. In the figure above two lines intersect. Which of the following must equal $180 - x$?

(A) $x + y$ (B) $x + z$ (C) $y + z$
(D) $y + w$ (E) $z + w$

31. If $s = 1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \frac{1}{32}$ and $t = 1 + \frac{1}{2}s$, then t exceeds s by

(A) $\frac{1}{4}$ (B) $\frac{1}{8}$ (C) $\frac{1}{16}$ (D) $\frac{1}{32}$ (E) $\frac{1}{64}$

Questions 32-33 refer to the following definition.

For all positive integers n,

let $(n) = \frac{1}{2}n$ if n is even;

let $(n) = 2n$ if n is odd.

32. If y is a prime number greater than 2, then $(y) =$

(A) $\frac{1}{4}y$ (B) $\frac{1}{2}y$ (C) y (D) $2y$ (E) $4y$

33. $(5) \cdot (10) =$

(A) (15) (B) (30) (C) (50)
(D) (100) (E) (200)



34. In the figure above, the radius of the smaller circle is half the radius of the larger circle. If the circles have the same center O, what is the ratio of the area of the shaded region to the area of the larger circle?

(A) 4:5 (B) 3:4 (C) 1:2
(D) 1:4 (E) 1:8

35. If the operation Δ is defined for all positive x and y by $x \Delta y = \frac{xy}{x + y}$, which of the following must be true for positive x, y, and z?

I. $x \Delta x = \frac{1}{2}x$

II. $x \Delta y = y \Delta x$

III. $x \Delta (y \Delta z) = (x \Delta y) \Delta z$

(A) I only (B) I and II only (C) I and III only
(D) II and III only (E) I, II, and III

IF YOU FINISH BEFORE TIME IS CALLED, YOU MAY CHECK YOUR WORK ON THIS SECTION ONLY. DO NOT WORK ON ANY OTHER SECTION IN THE TEST. **STOP**

How to Score the Sample Test

Before you can find out what your scores are on the College Board 200 to 800 scale, you need to determine your verbal, mathematical, and TSWE raw scores. The steps for doing so for each section of the test and a scoring worksheet are provided below and on page 62. Use the table on page 63 to determine your correct and incorrect answers for each section.

Determining Your Raw Scores

SAT-Verbal Sections 1 and 4

Step A: Count the number of correct answers for section 1 and record the number in the space provided on the worksheet. Then do the same for the incorrect answers. (Do not count omitted answers.) To determine subtotal A, use the formula:

$$\text{number correct} - \frac{\text{number incorrect}}{4} = \text{subtotal A}$$

Step B: Count the number of correct answers and the number of incorrect answers for section 4 and record the numbers in the spaces provided on the worksheet. To determine subtotal B, use the formula:

$$\text{number correct} - \frac{\text{number incorrect}}{4} = \text{subtotal B}$$

Step C: To obtain C, add subtotal A to subtotal B, keeping any decimals. Enter the resulting figure on the worksheet.

Step D: To obtain D, your raw verbal score, round C to the nearest whole number. (For example, any number from 44.50 to 45.49 rounds to 45.) Enter the resulting figure on the worksheet.

SAT-Mathematical Sections 2 and 5

Step A: Count the number of correct answers and the number of incorrect answers for section 2 and record the numbers in the spaces provided on the worksheet. To determine subtotal A, use the formula:

$$\text{number correct} - \frac{\text{number incorrect}}{4} = \text{subtotal A}$$

Step B: Count the number of correct answers and the number of incorrect answers for the *five-choice questions (questions 1 through 7 and 28 through 35)* in section 5 and record the numbers in the spaces provided on the worksheet. To determine subtotal B, use the formula:

$$\text{number correct} - \frac{\text{number incorrect}}{4} = \text{subtotal B}$$

Step C: Count the number of correct answers and the number of incorrect answers for the *four-choice questions (questions 8 through 27)* in section 5 and record the numbers in the spaces provided on the worksheet. To determine subtotal C, use the formula:

$$\text{number correct} - \frac{\text{number incorrect}}{3} = \text{subtotal C}$$

Note: Do not count any E responses to questions 8 through 27 as correct or incorrect. Because these four-choice questions have no E answer choices, E responses to these questions are treated as omits.

Step D: To obtain D, add subtotal A, subtotal B, and subtotal C, keeping any decimals. Enter the resulting figure on the worksheet.

Step E: To obtain E, your raw mathematical score, round D to the nearest whole number. (For example, any number from 44.50 to 45.49 rounds to 45.) Enter the resulting figure on the worksheet.

TSWE: Section 3

Step A: Count the number of correct answers for section 3 and record the number in the space provided on the worksheet. Then do the same for the incorrect answers. (Do not count omitted answers.) To determine your unrounded raw score, use the formula:

$$\text{number correct} - \frac{\text{number incorrect}}{4} = \text{total unrounded raw score}$$

Step B: To obtain B, your raw TSWE score, round A to the nearest whole number. (For example, any number from 34.50 to 35.49 rounds to 35.) Enter the resulting figure on the worksheet.

SCORING WORKSHEET FOR THE SAMPLE TEST

SAT-Verbal Sections

A. Section 1: _____ (no. correct) − 1/4 (_____ (no. incorrect)) = _____ subtotal A

B. Section 4: _____ (no. correct) − 1/4 (_____ (no. incorrect)) = _____ subtotal B

C. Total unrounded raw score (Total A + B) _____ C

D. Total rounded raw verbal score (Rounded to nearest whole number) _____ D

SAT-Mathematical Sections

A. Section 2: _____ (no. correct) − 1/4 (_____ (no. incorrect)) = _____ subtotal A

B. Section 5: Questions 1 through 7 and 28 through 35 (5-choice) _____ (no. correct) − 1/4 (_____ (no. incorrect)) = _____ subtotal B

C. Section 5: Questions 8 through 27 (4-choice) _____ (no. correct) − 1/3 (_____ (no. incorrect)) = _____ subtotal C

D. Total unrounded raw score (Total A + B + C) _____ D

E. Total rounded raw math score (Rounded to nearest whole number) _____ E

TSWE

A. Section 3: Total unrounded raw score _____ (no. correct) − 1/4 (_____ (no. incorrect)) = _____ A

B. Total rounded raw TSWE score (Rounded to nearest whole number) _____ B

ANSWERS TO SAMPLE TEST QUESTIONS AND PERCENTAGE OF STUDENTS ANSWERING EACH QUESTION CORRECTLY

Section 1—Verbal			Section 2—Mathematical			Section 3—TSWE			Section 4—Verbal			Section 5—Mathematical		
Question number	Correct answer	Percentage of students answering the question correctly	Question number	Correct answer	Percentage of students answering the question correctly	Question number	Correct answer	Percentage of students answering the question correctly	Question number	Correct answer	Percentage of students answering the question correctly	Question number	Correct answer	Percentage of students answering the question correctly
1	A	94%	1	D	78%	1	C	86%	1	B	89%	1	E	92%
2	C	97	2	B	93	2	B	91	2	B	83	2	B	75
3	B	89	3	D	88	3	B	81	3	D	76	3	B	79
4	D	87	4	B	87	4	A	84	4	D	65	4	A	69
5	B	80	5	B	64	5	C	87	5	A	64	5	A	70
6	E	63	6	A	81	6	E	90	6	D	41	6	B	62
7	E	66	7	C	76	7	D	77	7	D	34	7	C	70
8	D	42	8	B	82	8	C	78	8	A	35	8	C	80
9	C	49	9	B	75	9	C	71	9	B	25	9	A	80
10	B	36	10	E	64	10	E	78	10	B	24	10	A	82
11	B	30	11	C	70	11	B	82	11	E	90	11	B	74
12	B	27	12	E	43	12	A	74	12	C	74	12	A	80
13	E	32	13	C	65	13	D	73	13	D	84	13	D	59
14	A	20	14	D	63	14	A	91	14	B	22	14	C	78
15	A	24	15	A	44	15	D	50	15	C	18	15	A	59
16	D	79	16	C	40	16	E	68	16	A	89	16	B	75
17	A	64	17	E	39	17	B	60	17	C	84	17	A	71
18	D	31	18	D	37	18	C	57	18	A	72	18	D	26
19	D	32	19	C	22	19	B	65	19	B	62	19	B	60
20	C	40	20	C	23	20	D	52	20	D	49	20	C	56
21	C	47	21	A	18	21	E	79	21	E	37	21	A	83
22	A	87	22	E	22	22	A	44	22	B	35	22	B	36
23	E	83	23	C	18	23	C	62	23	A	35	23	C	38
24	B	68	24	A	18	24	C	39	24	B	22	24	D	22
25	E	47	25	E	14	25	B	67	25	E	17	25	A	56
26	B	79				26	B	94	26	E	77	26	D	25
27	C	55				27	C	68	27	A	49	27	D	14
28	E	69				28	D	83	28	D	46	28	A	56
29	A	68				29	A	79	29	A	62	29	B	44
30	D	49				30	A	50	30	E	23	30	A	32
31	C	35				31	E	65	31	C	61	31	E	27
32	B	36				32	D	63	32	D	80	32	D	45
33	D	31				33	A	82	33	B	61	33	D	16
34	C	20				34	C	69	34	A	64	34	B	16
35	B	18				35	B	90	35	D	60	35	E	10
36	A	88				36	C	54	36	C	69			
37	C	70				37	D	70	37	D	26			
38	D	71				38	E	40	38	B	44			
39	B	56				39	A	60	39	E	13			
40	E	28				40	D	44	40	A	50			
41	B	41				41	D	76						
42	D	31				42	E	70						
43	B	32				43	D	68						
44	B	22				44	B	59						
45	E	23				45	E	59						
						46	C	53						
						47	D	62						
						48	B	50						
						49	E	46						
						50	A	45						

Notes: The percentages for the SAT-verbal and SAT-mathematical sections are based on the analysis of the answer sheets for a random sample of juniors and seniors who took this test in May 1983 and whose mean scores were 438 on the SAT-verbal sections and 481 on the SAT-mathematical sections.

The percentages for TSWE are based on the analysis of the answer sheets for a random sample of all students who took this test in June 1981 and whose mean score was 43.

SCORE CONVERSION TABLE
Sample SAT and TSWE

	College Board Scaled Score			College Board Scaled Score			College Board Scaled Score
Raw Score	SAT-Verbal	SAT-Math	Raw Score	SAT-Verbal	SAT-Math	Raw Score	TSWE
85	800		40	460	610		
84	780		39	450	600		
83	770		38	450	590		
82	760		37	440	580		
81	750		36	430	570	50	60+
80	740		35	430	560	49	60+
79	730		34	420	550	48	60+
78	720		33	420	540	47	60+
77	710		32	410	530	46	59
76	700		31	400	520	45	58
75	690		30	400	510	44	57
74	680		29	390	500	43	56
73	670		28	380	490	42	55
72	660		27	380	480	41	54
71	660		26	370	470	40	53
70	650		25	360	460	39	52
69	640		24	360	450	38	51
68	630		23	350	450	37	50
67	630		22	340	440	36	49
66	620		21	340	430	35	48
65	610		20	330	420	34	47
64	600		19	320	410	33	46
63	600		18	320	400	32	45
62	590		17	310	390	31	44
61	580		16	300	380	30	43
60	580	800	15	290	380	29	42
59	570	780	14	290	370	28	41
58	570	770	13	280	360	27	40
57	560	760	12	270	350	26	39
56	550	750	11	270	340	25	38
55	550	740	10	260	340	24	37
54	540	740	9	250	330	23	36
53	530	730	8	240	320	22	35
52	530	720	7	240	310	21	34
51	520	710	6	230	310	20	33
50	520	700	5	220	300	19	32
49	510	690	4	220	290	18	31
48	510	680	3	210	280	17	30
47	500	670	2	200	280	16	29
46	490	660	1	200	270	15	28
45	490	660	0	200	260	14	27
44	480	650	−1	200	250	13	26
43	480	640	−2	200	250	12	25
42	470	630	−3	200	240	11	24
41	460	620	−4	200	230	10	22
			−5	200	220	9	21
			−6	200	210	8	20
			−7	200	210	or below	
			−8	200	200		
			or below				